Supplementary appendix

Cognitive assessment depth and future MoCA prediction in Parkinson's disease: an exploratory PPMI benchmark

This appendix records analysis provenance, feature definitions, evaluation rules and complete aggregate results. CSV tables retain full numerical precision; displayed values are rounded. The original models and predictions remain frozen.

Supplementary methods

Analysis sequence and scope

The original Clinical_hallmark analysis selected five genes and 11 Hallmark pathways from earlier transcriptomic work and tested them beyond core and enhanced clinical comparators. Its 36-month enhanced-plus-combined-RNA comparison remained primary. Subsequent individual feature, pathway, expanded-panel and GA studies were exploratory extensions. This paper synthesizes the matched clinical benchmark and reports the original RNA comparison for provenance. It does not merge unmatched plasma, Olink or earlier motor-prognosis studies into the same performance ranking. The internal source cohort was repeatedly analyzed. Of the 307 participants at 36 months, 292 had contributed to the earlier 558-person RNA discovery analysis and 233 were eligible for an earlier plasma cognitive analysis.

Eligibility and calendar rules

The source flow was 410 exact baseline RNA specimens, 387 with baseline timing/diagnosis/MoCA readiness, 354 after excluding 33 SWEDD participants, and 351 after requiring enhanced cognitive scores and forms. RNA phases contributed 250 and 101 participants. Among 351, unavailable timed MoCA records numbered 14, 28 and 42; selected-score quality holds numbered 4, 2 and 2; retained outcomes numbered 333, 321 and 307. The three windows were 9–15, 21–27 and 33–39 months. Selection preceded quality filtering and did not substitute a more favorable record. A 280-person common-outcome subset existed. Baseline sex was corroborated across baseline RNA, unified baseline and native screening information; this did not imply a later TRANS demographic visit occurred before index. Five missing diagnosis durations were retained for training-only imputation.

Missingness and preprocessing

Each numerical feature received its training median, then training mean and population-SD scaling. All-missing training columns were set to zero, and near-zero SD was replaced by one. A fixed missing indicator was retained for every numerical column. Categorical variables used complete fixed dictionaries, including unknown values. Test-form controls followed the selected parent cognitive test. No outcome imputation, future test forms, future medication values or withdrawal flags entered the predictors. The enhanced scores were complete by cohort construction. Medication status and examination state had no source variation. Optional predictor missingness was handled by the locked preprocessing; availability itself should not be interpreted as a biological effect.

Splits and penalties

One identity-component master allocation was inherited across horizons: five outer folds and three inner folds. Neither outcomes nor later follow-up availability determined that allocation. All learned preprocessing and selection were refitted within each inner training partition and the corresponding outer training partition. Ridge penalties were 0.001, 0.01, 0.1, 1 and 10; the intercept was not penalized. Ties within 10^{-12} favored the larger penalty. The earlier clinical comparators used pooled inner MAE; later fixed panels and R^2-oriented searches used pooled inner MSE. MSE normalization by the fixed outcome variance does not change candidate ranking. Calibration slope and intercept were descriptive regressions of observed outcomes on predictions, not target recalibration.

Search design and termination

The final v4 NSGA-II searches used population and offspring sizes of 1,024, a 10,000-generation ceiling, and crossover probability 0.9 and independent mutation probability 1/d per optional bit. Initial candidates included null, full, singleton, leave-one-out and reference panels, with remaining initialization random. Each of 30 arm/horizon/fold contexts had 20 main and five confirmation restarts. Each restart could visit at most 200,000 distinct masks. Stopping required at least 400 generations and 50,000 distinct masks, two consecutive 200-transition plateaus with progress below 10^{-5}, and selected-mask stability. All 25 restarts had to satisfy stagnation checks. Main and confirmation objective envelopes had to match the combined envelope at each attainable burden within 10^{-4} normalized MSE; at least three of five confirmation restarts had to approach the combined best within that tolerance. Matched random search could not outperform the GA envelope beyond the locked tolerance. Hard limits alone were not accepted as convergence. These conditions support empirical stabilization, not exhaustive coverage.

Search verification and audit limitation

Frozen verification checked 24,596,833 stored cache records and 368,952,555 recorded ridge solves across search and controls, 750 restart summaries and 60 outer models. These counts include repeated evaluations of the same modest participant sample and do not enlarge the independent sample size. Final predictions and metrics were independently reconstructed without model retraining. Completed restarts had discarded full per-generation trajectories. The audit checked recorded stopping summaries, minimum coverage and generation bounds, and main/confirmation/random envelopes, but could not replay every historical plateau. The original audit status was PASS_WITH_DOCUMENTED_AUDIT_LIMITATION. The integrated comparison independently verified 54 metric rows, 459 contrasts and 270 feature sets.

Earlier GA recipes

Earlier core-mandatory MAE and MSE searches used an accuracy anchor plus a one-standard-error allowance calculated from three inner-fold errors. Among eligible visited candidates they favored smaller acquisition burden. The later MoCA-mandatory optional-feature search and final v4 searches instead prioritized best pooled MSE, without that allowance. Their rows therefore compare complete model-selection recipes, not only mandatory-feature constraints. The unrestricted null mask meant intercept-only; the MoCA-mandatory null optional mask meant MoCA-only. Performance assigned to a recipe cannot be assigned retrospectively to a consensus feature subset.

Temporal evaluation

The temporal candidate frame contained 570 participants; 227 unique participants were eligible at one or more horizons, and 33 had outcomes at all three. There were 222, 182 and 43 calendar-selected outcomes; score checks retained 221, 182 and 43; withdrawal holds and comparator availability led to final counts of 213, 175 and 39. Index years were 2023–2025. Non-MoCA sources were frozen on 20 October 2026 and MoCA was refreshed on 28 October before any temporal scoring. The amended audit added seven valid pairs and two participants without changing earlier bound outcomes. Historical diagnosis-date fallback and age rules were frozen before scoring. No overlaps with source clinical training/test identities were found; 224 of 227 had earlier motor-model exposure. The clinical models were equal-weight ensembles of five earlier MAE-tuned outer models. Typical change used the average of the five stored source-fold median offsets. No temporal data were used for fitting or recalibration. The 28 October MoCA amendment preceded temporal scoring; it did not create an external cohort or remove prior motor-analysis exposure.

Temporal sensitivity interpretation

Complete-core samples were 204, 171 and 38; age-resolution exclusions left 207, 171 and 39. Other frozen sensitivities used stricter provenance or the common 33-person follow-up subset. The age-excluded 36-month result is an explicit alias of the main 36-month result, not a new independent analysis. Median elapsed follow-up was 13, 25 and 36 months, and median MoCA change was zero at each horizon. Absence of a timed outcome does not by itself establish dropout, death or a complete opportunity for follow-up. The full sensitivity metrics

and intervals are in Table S6. The 24-month age-resolution exclusion gave enhanced-minus-core MAE −0.141 (95% interval −0.273 to −0.005; n=171). Its enhanced-minus-persistence interval crossed zero. This exploratory sensitivity does not supersede the uncertain main 24-month contrast.

Preserved RNA methods

Raw counts were transformed within sample as $\log_2(1 + 10^6 \times \text{count} / \text{total count across all 58,780 genes})$. The fixed genes were LINC01806, CISD2, CDK6, CPT1A and NPIPB8. A training gene required at least 10 counts in at least max(3, ceiling[0.20 × training n]) samples and nonconstant log expression. Training-standardized eligible member expression was averaged for each fixed Hallmark set, then the pathway score was standardized using training information. At least five eligible members were required. Failed features remained fixed zero columns without replacement. A block-penalized ridge used the clinical penalty grid above and RNA penalties 0.01, 0.1, 1 and 10. Each augmented arm also offered all five clinical-only candidates as genuine RNA-off options. Selection used pooled inner MAE; ties preferred RNA-off, then larger RNA and clinical penalties. This allowed an augmented arm to return the clinical prediction exactly. Zero-width intervals in RNA-off results are not biological equivalence.

Uncertainty and interpretation

The internal comparison supplied every pair among 18 recipes at each horizon, giving 459 contrasts. Bootstrap resampling was paired by participant identity with 5,000 replicates. Contrast direction in CSV is model a minus model b; lower MAE and higher R^2 are preferable. These are conditional intervals from frozen predictions, without multiplicity adjustment or a full rerun of fitting and selection. No noninferiority or equivalence margin was prespecified. The main paper does not claim that maximum observed R^2 estimates performance on untouched data. This manuscript package was generated solely from aggregate evidence; it did not retrain, rescore participants or issue paid queries. Only one master fold partition was used; no repeated-outer-cross-validation uncertainty is represented. The conditional intervals also omit the broader adaptive sequence of study choices.

Table S1 Cohort flow and follow-up availability

Stage	Source	Temporal
Initial audited frame	410 RNA participants	570 later candidates
Baseline eligible frame	351	227 eligible at ≥1 horizon
12-month outcome	333	213
24-month outcome	321	175
36-month outcome	307	39
All three outcomes	280	33
Index years	2010–2016	2023–2025

Table S2 Predictor definitions

Substantive predictor	Panel membership	Acquisition group
Age	Core	demographic history
Sex	Core	demographic history

Substantive predictor	Panel membership	Acquisition group
Education	Core	demographic history
Baseline MoCA	Core; compact 4	MoCA
Diagnosis duration	Core	PD history
MDS-UPDRS II	Core	MDS UPDRS II
MDS-UPDRS III	Core	MDS UPDRS III
GDS-15	Core	GDS15
Medication status	Core	medication history
Motor examination state	Core	MDS UPDRS III
SDMT	Optional 18; compact 4	SDMT
HVLT immediate recall	Optional 18; compact 4	HVLT
HVLT delayed recall	Optional 18; compact 4	HVLT
Animal fluency	Optional 18	Animal fluency
Letter-number sequencing	Optional 18	LNS
Judgment of line orientation	Optional 18	JLO
HVLT recognition discrimination	Optional 18	HVLT
REM sleep behaviour symptoms	Optional 18	RBDSQ
Autonomic gastrointestinal	Optional 18	SCOPA AUT
Autonomic urinary	Optional 18	SCOPA AUT
Autonomic cardiovascular	Optional 18	SCOPA AUT
Postural systolic BP drop	Optional 18	postural BP
Postural diastolic BP drop	Optional 18	postural BP
Trait anxiety	Optional 18	STAI
Apathy	Optional 18	MDS UPDRS I
Subjective cognitive complaint	Optional 18	MDS UPDRS I
APOE ε4 dosage	Optional 18	APOE assay
Neutrophil-to-lymphocyte ratio	Optional 18	CBC differential

APOE ε4 dosage is the allele count. NLR is the neutrophil-to-lymphocyte ratio from the historical CBC. Autonomic domains are gastrointestinal, urinary and cardiovascular components; blood-pressure drops compare postural measurements. Apathy and subjective cognitive complaint are baseline MDS-UPDRS I items. These are the frozen feature bindings, not newly selected thresholds. Core and enhanced scores are baseline variables; future values are outcomes or audit information only. Full encoded columns include the attached form controls and missing indicators.

Table S3 Complete internal model metrics

The accompanying Table_S3_All_54_Internal_Metrics.csv contains all 18 recipes at all three horizons, including R^2, MSE, RMSE, MAE, mean prediction error, calibration slope/intercept and out-of-range prediction fraction. Figure S1 displays their R^2 estimates. Table 2 gives the main reference models. The additional rows include optional-18-only, earlier GA recipes and matched final random-search controls.

Table S4 All paired internal contrasts

Table_S4_All_459_Paired_Contrasts.csv supplies every comparison with its horizon, participant count, difference, conditional interval, bootstrap count and seed. The difference is a minus b. Figures and text use explicitly oriented contrasts; reversing a contrast negates its estimate and reverses/negates its interval endpoints.

Table S5 Selection frequencies

Table_S5_Feature_Selection_Frequencies.csv records selection counts by model, feature and horizon. Table_S5b_All_270_Feature_Sets.csv lists the exact fold-specific sets and acquisition groups. Figure S2 displays the unrestricted final GA. Counts measure inclusion among five outer models and are not causal effects, coefficient magnitudes or evidence that a feature would be selected in another cohort.

Table S6 Temporal evaluation and sensitivities

Evaluation	n	ΔMAE	95% conditional interval
main 12	213	-0.069	-0.195 to 0.058
complete core 12	204	-0.053	-0.184 to 0.078
strict inputs 12	213	-0.089	-0.219 to 0.039
exclude age resolved 12	207	-0.069	-0.196 to 0.057
common horizons 12	33	-0.082	-0.401 to 0.232
main 24	175	-0.128	-0.262 to 0.004
complete core 24	171	-0.118	-0.245 to 0.008
strict inputs 24	175	-0.127	-0.254 to 0.002
exclude age resolved 24	171	-0.141	-0.273 to -0.005
common horizons 24	33	-0.106	-0.460 to 0.237
main 36	39	-0.012	-0.334 to 0.304
complete core 36	38	0.045	-0.254 to 0.348
strict inputs 36	39	-0.045	-0.335 to 0.259
exclude age resolved 36	39	-0.012	-0.334 to 0.304
common horizons 36	33	0.004	-0.339 to 0.339

Differences are enhanced minus core. The full model-specific metrics, calibration summaries and conditional intervals are in Table_S6_Temporal_Metrics_All_Sensitivities.csv. The age-excluded 36-month row reuses the main result by design.

Table S7 Original RNA contrasts

The 11 fixed Hallmark sets were apical junction, apoptosis, coagulation, complement, heme metabolism, hypoxia, IL6–JAK–STAT3 signaling, inflammatory response, interferon alpha response, interferon gamma response and PI3K–AKT–mTOR signaling. Full frozen membership and gene-version bindings remain in the original analysis contract; no pathways were selected using cognitive outcomes.

Horizon and comparator	RNA block	ΔMAE	95% conditional interval	Off folds
12 core	Genes5	0.0168	-0.0127 to 0.0480	1
12 core	Hallmark11	-0.0027	-0.0178 to 0.0121	1
12 core	Combined	0.0059	-0.0222 to 0.0329	1
12 enhanced	Genes5	0.0076	-0.0266 to 0.0424	0
12 enhanced	Hallmark11	0.0005	-0.0066 to 0.0074	2
12 enhanced	Combined	0.0055	-0.0146 to 0.0264	0
24 core	Genes5	0.0065	-0.0033 to 0.0166	3
24 core	Hallmark11	0.0144	-0.0223 to 0.0533	2
24 core	Combined	0.0345	-0.0114 to 0.0824	2
24 enhanced	Genes5	0.0026	-0.0088 to 0.0140	3
24 enhanced	Hallmark11	0.0108	-0.0137 to 0.0376	2
24 enhanced	Combined	0.0125	-0.0176 to 0.0429	2
36 core	Genes5	0.0000	0.0000 to 0.0000	5
36 core	Hallmark11	0.0059	-0.0076 to 0.0192	4
36 core	Combined	0.0071	-0.0080 to 0.0227	4
36 enhanced	Genes5	0.0000	0.0000 to 0.0000	5
36 enhanced	Hallmark11	0.0077	-0.0019 to 0.0168	4
36 enhanced	Combined	0.0028	-0.0012 to 0.0067	4

Differences are augmented minus clinical-only MAE. Combined RNA with the enhanced comparator at 36 months was the original primary comparison. Off folds counts selections of the clinical-only option among five outer folds. The original seed-specific intervals are retained; they need not equal intervals recomputed under a later unified comparison seed. Full precision is retained in Table_S7_Original_RNA_Contrasts.csv.

Reporting map and remaining author declarations

Data and participants

Main Methods and Table 1; Supplementary Table S1. This is a secondary retrospective PPMI analysis with explicit prior use and timing limitations.

Outcome and predictors

Main Methods and Supplementary Table S2. Outcome is continuous future MoCA; predictor selection is entirely baseline-bound.

Model development and evaluation

Main Methods and supplementary search, preprocessing and temporal sections. Fixed penalties, grouped folds, inner-only selection and separate temporal ensembles are described.

Results and uncertainty

Main Tables 1–2 and Figures 1–2; complete Tables S3–S10. Conditional, unadjusted intervals and small temporal samples are identified.

Implementation and availability

Main declarations require the author-approved data/code access statement; restricted source records are not public artifacts. No deployment-ready model is claimed.

Author completion

Names, affiliations, funding, conflicts, contributions, local ethics determination, current PPMI acknowledgement, related-work disclosure and final AI-use responsibility statement remain for author completion. This map supports reporting review and is not a completed official TRIPOD+AI checklist.

Supplementary figure captions

Figure S1. R^2 for all 18 frozen internal model recipes at three horizons. Samples and folds are matched within each horizon. Points show pooled out-of-fold estimates without displaying intervals; complete metrics and paired intervals are in Tables S3 and S4. MAE- and MSE-tuned recipes are labelled separately. This is an exploratory comparison in previously analyzed participants. Figures were generated from aggregate results using reproducible AI-assisted plotting code.

Figure S2. Feature selection across the five outer folds of the final unrestricted GA. Cell values range from zero to five. Each feature is counted once per outer fold, regardless of the number of encoded columns it controls. The display does not define a new fitted consensus model or establish feature importance. The source cohort had no medication/state variation. Plotting used frozen aggregate counts and AI-assisted code.

Additional reporting and interpretation details

Table S8 Distinct analysis questions and original primary contrasts

Analysis stage	Designated question	Result and interpretation
Original RNA study	36-month enhanced + five genes/11 Hallmarks minus enhanced MAE	+0.002754 (−0.001184, 0.006681). No demonstrated benefit for this tested representation.
Final v4 GA study	36-month unrestricted GA minus MoCA-mandatory GA R^2	0.000; conditional interval zero within floating-point precision. No observed gain from relaxing mandatory MoCA.
Temporal clinical evaluation	12-month enhanced minus core MAE	−0.069 (−0.195, 0.058). Earlier clinical ensembles only; no compact-panel or final-GA temporal evaluation.
Current clinical emphasis	Matched core, enhanced, compact and expanded comparisons at all three horizons	Exploratory integration after prior source-outcome use. No new primary hypothesis or confirmatory validation is assigned.

Separate contracts defined these questions before their own runs. This does not imply that the broader research program or current reporting emphasis was prospectively specified. Numerical near-zero GA intervals reflect identical evaluated predictions; they are not a population equivalence test. The GA contrast shown uses the unified comparison records; the original GA bootstrap seed differs, but the zero result is unchanged.

Table S9 Descriptive internal calibration

Panel	Months	Slope	Intercept	Mean error
Four cognitive scores	12	1.028	-0.736	0.007
Four cognitive scores	24	1.032	-0.836	-0.003
Four cognitive scores	36	1.033	-0.873	-0.011
Expanded 28 predictors	12	1.250	-6.588	0.005
Expanded 28 predictors	24	1.251	-6.588	-0.011
Expanded 28 predictors	36	1.233	-6.150	-0.004

Calibration regresses observed scores on fixed out-of-fold predictions; ideal slope/intercept are one/zero. Mean error is predicted minus observed. Values are descriptive, with no calibration uncertainty intervals or target recalibration. They cannot establish comparative calibration superiority. Full unrounded results remain in Table S3.

Table S10 Tuning criterion and paired enhanced-minus-core error

Inner tuning	Months	MAE difference	95% conditional interval
MSE	12	-0.094	-0.179 to -0.013
MSE	24	-0.116	-0.220 to -0.015
MSE	36	-0.144	-0.230 to -0.057
MAE	12	-0.114	-0.199 to -0.033
MAE	24	-0.086	-0.184 to 0.013
MAE	36	-0.176	-0.271 to -0.078

Negative values favor enhanced assessment. Both loss functions select a ridge penalty on the same inner validation participants, but need not choose the same prediction rule. These intervals use the unified comparison seed, retain all horizons and are unadjusted. The MAE-tuned 24-month interval crosses zero. Figure 2A shows the MSE-tuned comparison; it is not evidence of tuning-independent benefit.

Target, generalization and clinical utility

The target is future observed MoCA in this eligibility-defined frame. It is not a causal treatment effect, a competing-risk cognitive outcome or a dementia diagnosis. No formal clinical utility threshold, noninferiority margin, cost study or decision intervention was evaluated. Source eligibility depended on RNA and enhanced-test availability even for clinical-only comparisons. Site transportability and performance of the four-score or final GA candidates in a new population remain untested. No additional participant scoring was performed for this revision.